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Constant Acceleration in 2D

The same SUVAT, but every letter is a vector. Do the $\mathbf{i}$ part and the $\mathbf{j}$ part separately — then speed and direction come from Pythagoras and $\tan$.

THE TOPIC ON ONE MAP

Constant Acceleration in 2D

01 — Velocity vectors

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\text{speed} = |\mathbf{v}|$$

$\mathbf{a}$ from two velocities

02 — Direction of motion

parallel to $\mathbf{a}$ vector

bearing

same speed

03 — Position vectors

$$\mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

distance from O

VECTOR SUVAT • two equations you need

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\text{speed} = \sqrt{v_i^2 + v_j^2}$$

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

$$\text{direction: } \tan \theta = \frac{v_j}{v_i} \text{ with a sketch}$$

01 Velocity Vectors: $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

Everything is a vector: add and multiply the **i** parts together and the **j** parts together. Speed is the magnitude of the velocity; acceleration is the change of velocity divided by the time.

WHEN YOU SEE → YOU DO

“find the speed”

$$\rightarrow |\mathbf{v}| = \sqrt{a^2 + b^2}$$

velocities at two times

$$\rightarrow \mathbf{a} = \frac{\mathbf{v}_2 - \mathbf{v}_1}{t_2 - t_1}$$

“ $\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$ ”→ the bracket times t is **a**

“both boats have the same speed”

$$\rightarrow |\mathbf{v}_A|^2 = |\mathbf{v}_B|^2 - \text{a quadratic in } t$$

km h⁻¹, “hours after noon”→ keep hours; $t = 0$ at noon

WORKED MODEL • JUNE 2022

$\mathbf{v} = -\mathbf{i} + 4\mathbf{j}$ at $t = 1$ and $5\mathbf{i} - 8\mathbf{j}$ at $t = 4$. Find the speed at $t = 3.5$.

1 $\mathbf{a} = \frac{(6\mathbf{i} - 12\mathbf{j})}{3} = 2\mathbf{i} - 4\mathbf{j}$

change ÷ time

2 $\mathbf{v} = -\mathbf{i} + 4\mathbf{j} + 2.5(2\mathbf{i} - 4\mathbf{j}) = 4\mathbf{i} - 6\mathbf{j}$

2.5 s after $t = 1$

ANSWER $\sqrt{16 + 36} = \sqrt{52} \approx 7.2 \text{ m s}^{-1}$

IDEA 01

Practise: Velocity Vectors: $v = u + at$

My questions first, then real past papers.

Q1 DR ESLAM Velocity at a time

M1 • TOPIC 5 3 marks

A particle has initial velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$ and acceleration $(\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. Find its velocity and speed at $t = 4$.

Q2 DR ESLAM Acceleration from two velocities

M1 • TOPIC 5 3 marks

The velocity changes from $(3\mathbf{i} + \mathbf{j})$ to $(9\mathbf{i} - 7\mathbf{j}) \text{ m s}^{-1}$ in 2 s. Find the acceleration and its magnitude.

$\mathbf{a} = \frac{\mathbf{v} - \mathbf{u}}{t}$ — do $\mathbf{i}$ and $\mathbf{j}$ separately.

PAST PAPERS

Q3 **PAST PAPER** Pearson IAL WME01 • June 2022 • Q6**M1 • TOPIC 5** **6 marks**

6. A particle P is moving with constant acceleration.

At time $t = 1$ second, P has velocity $(-\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$

At time $t = 4$ seconds, P has velocity $(5\mathbf{i} - 8\mathbf{j}) \text{ m s}^{-1}$

Find the speed of P at time $t = 3.5$ seconds.

(6)

Q4 PAST PAPER Pearson IAL WME01 · June 2025 · Q7

M1 · TOPIC 5 13 marks

7. [In this question **i** and **j** are horizontal unit vectors.]Two boats, *A* and *B*, are each moving with constant acceleration.At time *t* hours after noon, boat *A* has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

(a) Find the magnitude of the acceleration of *A*.

(2)

When $t = 2$, the velocity of *B* is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$ When $t = 5$, the velocity of *B* is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$ (b) Find the acceleration of *B*, giving your answer in terms of **i** and **j**.

(2)

(c) Find the velocity of *B* at time $t = 0$, giving your answer in terms of **i** and **j**.

(2)

At time T_1 hours after noon, both boats are moving with the same speed.(d) Find the exact value of T_1

(4)

At time T_2 hours after noon, both boats are moving in the same direction.(e) Show that $3T_2^2 + pT_2 + q = 0$, where *p* and *q* are integers to be found.

(3)

02 Direction of Motion and Bearings

The direction of motion is the direction of the **velocity**. “Parallel to $p\mathbf{i} + q\mathbf{j}$ ” means the parts of $\mathbf{v}$ are in the ratio $p : q$. For a bearing, sketch the vector from north and measure clockwise.

WHEN YOU SEE → YOU DO

“moving in the direction of $(-\mathbf{i} + 2\mathbf{j})$ ”

→ $\frac{v_j}{v_i} = \frac{2}{-1}$, keep the signs

“moving due east / parallel to $\mathbf{i}$ ”

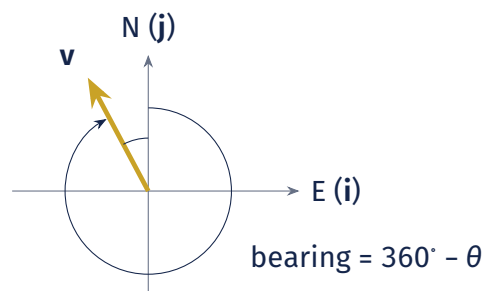
→ $\mathbf{j}$ part = 0

“as a bearing”

→ sketch; angle clockwise from north

“the velocity of P equals the velocity of Q ”

→ equal $\mathbf{i}$ parts and equal $\mathbf{j}$ parts



WORKED MODEL • JUNE 2021

$\mathbf{v} = (3 - 4T)\mathbf{i} + (5 + 4T)\mathbf{j}$ is parallel to $(-\mathbf{i} + 2\mathbf{j})$. Find T .

1 $5 + 4T = -2(3 - 4T)$

$\mathbf{j}$ part = $-2 \times \mathbf{i}$ part

ANSWER $T = \frac{11}{4}$ hours

× **TRAP** Using the position vector for the direction of motion. Direction of motion = direction of $\mathbf{v}$.

IDEA 02

Practise: Direction of Motion and Bearings

My questions first, then real past papers.

Q5 DR ESLAM Bearing of a velocity

M1 • TOPIC 5 2 marks

$\mathbf{i}$ is due east and $\mathbf{j}$ due north. Find the bearing of the velocity $(-3\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$.

Q6 DR ESLAM Parallel to a vector

M1 • TOPIC 5 4 marks

$\mathbf{v} = (2 + t)\mathbf{i} + (5 - 2t)\mathbf{j}$. Find when the particle moves (a) parallel to $\mathbf{i}$ (b) parallel to $\mathbf{i} + \mathbf{j}$.

Parallel to $\mathbf{i}$: the $\mathbf{j}$ part is 0. Parallel to $\mathbf{i} + \mathbf{j}$: the two parts are equal.

PAST PAPERS

Q7 PAST PAPER Pearson IAL WME01 • June 2021 • Q5

M1 • TOPIC 5 9 marks

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

A particle P is moving with constant acceleration. At 2 pm, the velocity of P is $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$ and at 2.30 pm the velocity of P is $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time T hours after 2 pm, P is moving in the direction of the vector $(-\mathbf{i} + 2\mathbf{j})$

(a) Find the value of T .

(6)

Another particle, Q , has velocity $\mathbf{v}_Q \text{ km h}^{-1}$ at time t hours after 2 pm, where

$$\mathbf{v}_Q = (-4 - 2t)\mathbf{i} + (\mu + 3t)\mathbf{j}$$

and μ is a constant.

Given that there is an instant when the velocity of P is equal to the velocity of Q ,

(b) find the value of μ .

(3)

Q8**PAST PAPER**

Pearson IAL WME01 • October 2023 • Q4

M1 • TOPIC 5**10 marks**

4. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively.]

A particle P moves with constant acceleration $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})\text{ m s}^{-2}$, where λ is a positive constant.

At time $t = 0$, the velocity of P is $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

- (a) Find the velocity of P when $t = 5\text{ s}$, giving your answer in terms of $\mathbf{i}$, $\mathbf{j}$ and λ .

(2)

The speed of P when $t = 5\text{ s}$ is 13 m s^{-1}

- (b) Show that

$$25\lambda^2 - 42\lambda - 16 = 0$$

(3)

- (c) Find the direction of motion of P when $t = 4\text{ s}$, giving your answer as a bearing to the nearest degree.

(5)

03 Position Vectors: $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

Start from where the particle is at $t = 0$, add $\mathbf{u}t$, add $\frac{1}{2}\mathbf{a}t^2$. The distance from O is the magnitude of $\mathbf{r}$; the distance between two particles is the magnitude of the difference.

WHEN YOU SEE → YOU DO

“the position vector at time t ” $\longrightarrow \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

“starts from rest at O ” $\longrightarrow \mathbf{r}_0 = \mathbf{0}, \mathbf{u} = \mathbf{0}$

“distance from O ” $\longrightarrow |\mathbf{r}|$

“due north of O ” $\longrightarrow \mathbf{i} \text{ part} = 0$

WORKED MODEL • A PARTICLE FROM REST

From rest at O with $\mathbf{a} = 2\mathbf{i} - \mathbf{j}$. Distance from O at $t = 3$?

1 $\mathbf{r} = \frac{1}{2}(2\mathbf{i} - \mathbf{j})(9) = 9\mathbf{i} - 4.5\mathbf{j}$

$\mathbf{r}_0 = \mathbf{u} = \mathbf{0}$

ANSWER $|\mathbf{r}| = \sqrt{81 + 20.25} \approx 10.1 \text{ m}$

IDEA 03

Practise: Position Vectors: $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

Questions written to practise exactly this idea.

Q9**DR ESLAM**

Where is it?

M1 • TOPIC 5 3 marks

$\mathbf{r}_0 = 2\mathbf{i} + \mathbf{j}$, $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j}$, $\mathbf{a} = \mathbf{i} + \mathbf{j}$. Find the position vector at $t = 2$.

Q10**DR ESLAM**

Distance from the start

M1 • TOPIC 5 3 marks

A particle starts from rest at O with acceleration $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-2}$. Find its distance from O after 3 s.

ALL The Whole Topic on One Page

Tear this page out before the exam.

01 Velocity

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t; \text{ speed} = |\mathbf{v}|.$$

02 Direction

Parts of $\mathbf{v}$ in the ratio of the vector; bearing from a sketch.

03 Position

$$\mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2; \text{ distance} = |\mathbf{r}|.$$

BOTTOM LINE $\mathbf{i}$ with $\mathbf{i}$, $\mathbf{j}$ with $\mathbf{j}$ • speed by Pythagoras • direction from $\mathbf{v}$, never from $\mathbf{r}$.

Answers

Constant Acceleration in 2D • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • VELOCITY VECTORS: $\mathbf{v} = \mathbf{u} + \mathbf{at}$

- Q1** Dr Eslam • Velocity at a time $\mathbf{v} = 6\mathbf{i} + 5\mathbf{j}$; speed $\sqrt{61} \approx 7.81 \text{ m s}^{-1}$
- Q2** Dr Eslam • Acceleration from two velocities $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$; $|\mathbf{a}| = 5 \text{ m s}^{-2}$
- Q3** Pearson IAL WME01 • June 2022 • Q6 $\mathbf{a} = 2\mathbf{i} - 4\mathbf{j}$; $\mathbf{v}(3.5) = 4\mathbf{i} - 6\mathbf{j}$; speed $\sqrt{52} = 2\sqrt{13} \approx 7.2 \text{ m s}^{-1}$
- Q4** Pearson IAL WME01 • June 2025 • Q7 (a) $\sqrt{17} \approx 4.1 \text{ km h}^{-2}$ (b) $-\mathbf{i} - 2\mathbf{j}$ (c) $6\mathbf{i} + 5\mathbf{j}$ (d) $T_1 = \frac{-1 + \sqrt{17}}{2}$ (e) $p = -14, q = 4$

IDEA 02 • DIRECTION OF MOTION AND BEARINGS

- Q5** Dr Eslam • Bearing of a velocity north-west: 315°
- Q6** Dr Eslam • Parallel to a vector (a) $t = 2.5$ (b) $2 + t = 5 - 2t \Rightarrow t = 1$
- Q7** Pearson IAL WME01 • June 2021 • Q5 (a) $T = \frac{11}{4} \text{ h}$ (b) $\mu = 8.5$
- Q8** Pearson IAL WME01 • October 2023 • Q4 (a) $(5 - 5\lambda)\mathbf{i} + (-8 + 10\lambda)\mathbf{j}$ (b) $25\lambda^2 - 42\lambda - 16 = 0$ (c) 339°

IDEA 03 • POSITION VECTORS: $\mathbf{r} = \mathbf{r}_0 + \mathbf{ut} + \frac{1}{2}\mathbf{at}^2$

- Q9** Dr Eslam • Where is it? $\mathbf{r} = 10\mathbf{i} - \mathbf{j}$
- Q10** Dr Eslam • Distance from the start $\mathbf{r} = 9\mathbf{i} - 4.5\mathbf{j}$; $\sqrt{101.25} \approx 10.1 \text{ m}$